DHOLE PATIL COLLEGE OF ENGINEERING, PUNE

DSBDA LAB MANUAL

(Department of Computer Engineering)

Practical 1

AIM:

Data Wrangling, I
Perform the following operations using Python on any open source dataset (e.g., data.csv)
1. Import all the required Python Libraries.
2. Locate an open source data from the web (e.g. https://www.kaggle.com). Provide a clear
description of the data and its source (i.e., URL of the web site).
3. Load the Dataset into pandas data frame.
4. Data Preprocessing: check for missing values in the data using pandas insult(), describe()
function to get some initial statistics. Provide variable descriptions. Types of variables etc. Check the dimensions of the data frame.
5. Data Formatting and Data Normalization: Summarize the types of variables by checking
the data types (i.e., character, numeric, integer, factor, and logical) of the variables in the data set. If variables are not in the correct data type, apply proper type conversions.
6. Turn categorical variables into quantitative variables in Python.

REQUIREMENT:

AnacondaInstaller

Windows10OS

JupyterNotebook

THEORY:

Data Wrangling:

- Data wrangling is the process of **cleaning and organizing messy data**
- It converts raw data into a **usable format for analysis**
- Involves **transforming and mapping data** from one form to another
- Helps in handling **large and complex datasets**

Goals of Data Wrangling:

- Collect data from **multiple sources** for better insights
- Provide **accurate and useful data** for analysis
- Reduce time spent on **data cleaning and preparation**
- Allow analysts to focus more on **data analysis**
- Support **better decision-making** in organizations
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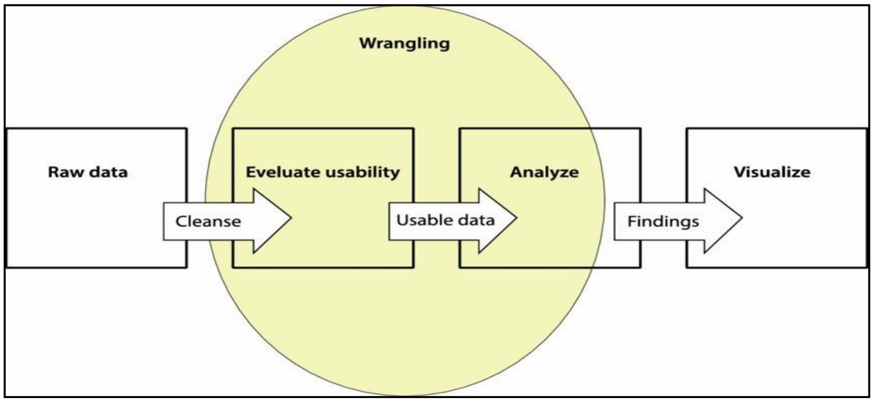
Key steps to DataWrangling:



LibrariesUsed:

Pandas

Numpy

Input:

[NOTE: Students need to paste their own Code here which they have performed]

Output:

[NOTE: Students need to paste the Screenshots of the Output of their code own Code here which they have performed, Individual Students Name should be there in the Screenshot]

CONCLUSION:

This experiment demonstrated basic data wrangling in Python, including loading a dataset, checking missing values, analyzing data types, and converting categorical variables into numerical form for further analysis.

Practical 2

AIM:

DataWranglingII

Create an "Academic performance" dataset of students and perform the following operations using Python.
1. Scan all variables for missing values and inconsistencies. If there are missing values and/or inconsistencies, use any of the suitable techniques to deal with them.
2. Scan all numeric variables for outliers. If there are outliers, use any of the suitable techniques to deal with them.
3. Apply data transformations on at least one of the variables. The purpose of this transformation should be one of the following reasons: to change the scale for better understanding of the variable, to convert a non-linear relation into a linear one, or to decrease the skewness and convert the distribution into a normal distribution.

THEORY:

Data Wrangling:

- Data wrangling is the process of **cleaning, organizing, and transforming raw data**
- Converts data into a **useful format for analysis and decision-making**
- Also known as **data cleaning or data munging**
- Helps handle **complex data efficiently**
- Methods vary based on **project goals and data type**

Benefits of Data Wrangling:

- Improves **data usability and quality**
- Converts data into a **compatible format**
- Helps in **quick data processing and automation**
- Integrates data from **multiple sources** (databases, files, APIs)
- Allows handling of **large volumes of data**
- Makes data sharing easier

Data Wrangling Tools:

- **Excel / Power Query**
- **Tabula**
- **Google DataPrep**

CONCLUSION:

This experiment demonstrated data wrangling techniques such as handling missing values, detecting and treating outliers, and applying data transformation to improve the dataset for analysis.

Practical 3

AIM:

Provide summery statistics for a dataset with numeric varibles grouped by one of the qualitative variables.

write a python program to display some basic statistical details like percentile, mean, standard derivation , etc.

REQUIREMENT:

Anaconda Installer

Windows10OS

JupyterNotebook

THEORY:

Step 1: Summary Statistics

Provide summary statistics such as **mean, median, mode, and standard deviation**.

Mean:

- Mean is the **average value** of the dataset
- Formula:
  Mean = (Sum of all values) / (Total number of values)

Median:

- Median is the **middle value** of a **sorted dataset**
- Arrange data in **ascending order** before finding median

Mode:

- Mode is the **most frequently occurring value** in the dataset

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Example:

Data: 36, 40, 32, 42, 30

- **Mean = (36 + 40 + 32 + 42 + 30) / 5 = 36 kg**
- Arrange data: 30, 32, 36, 40, 42
- **Median = 36 kg**
- **Mode = No mode (no value repeats)**

Standard Deviation:

- Measures how much data values **deviate from the mean**
- It is the **square root of variance**
- Helps to understand **data spread or variation**

Standard Deviation Formula:

$$\sigma = \sqrt{\frac{(x_i - \mu)^2}{N}}$$

Where:

- σ = Standard Deviation
- x_i = Each data value
- μ = Mean of the dataset
- N = Total number of values

LibrariesUsed:

Pandas

Numpy

Seaborn

CONCLUSION:

From this experiment, we learned how to calculate **mean, median, and mode** to understand the central tendency of data. These measures help in summarizing data and analyzing its distribution effectively..

Practical 4

AIM:

Create a linear regression model using python to predict home prise using boston housing dataset**.**

REQUIREMENT:

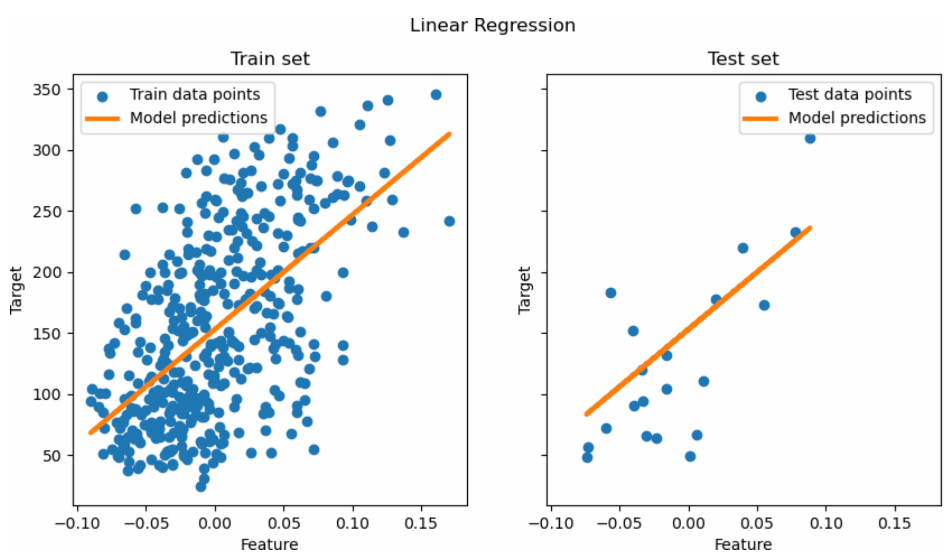
AnacondaInstaller

Windows10OS

JupyterNotebook

THEORY:

Linear Regression in data science:



The term regression is used when you try to find the relationship between variables. In Machine Learning and in statistical modeling, that relationship is used to predict the outcome of events.

SimpleLinearRegression:

Simple linear regression is an approach for predicting a response using a single feature. It is assumed that the two variables are linearly related. Hence, we try to find a linear function that predicts the response value(y) as accurately as possible as afunction of the feature or independent variable(x).

Multiple linear regression:

Multiple linear regressionattempts to model the relationshipbetween two ormore featuresandaresponsebyfittingalinearequationtotheobserveddata. Clearly, it is nothing but an extension of simple linear regression.

APPLICATIONS:

Trend lines: Trend lines show how data changes over time and can be predicted using linear regression.

Finance: In finance, linear regression helps analyze investment risk, and in biology, it is used to study relationships between variables.

Libraries Uesd:

Pandas

Sklearn

CONCLUSION:

In this experiment we have studied abbout linear regression and done houseprise prediction using boston housing dataset.

Practical 5

AIM:

Implementlogisticregressionusingpythontoperformclassifiactiononsocial network_ads , cv dataset.

REQUIREMENT:

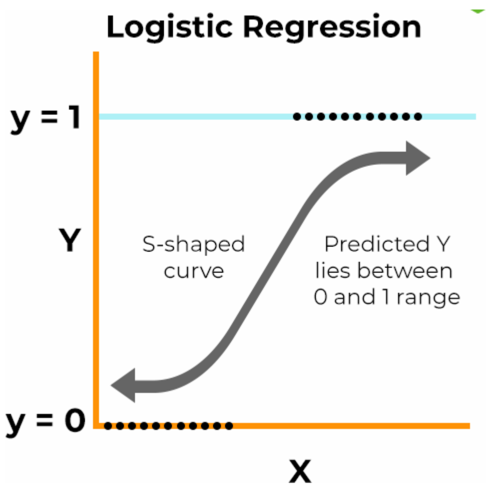
AnacondaInstaller

Windows10OS

JupyterNotebook

THEORY:

Logistic Regression



Logistic regression is a supervised learning algorithm used for classification problems where the output is categorical (such as yes/no or 0/1). It helps in categorizing data based on different features like color, size, shape, and weight. For example, it can classify a dog as an animal and an orange as not an animal. In binary logistic regression, there are only two possible outcomes, while multinomial logistic regression deals with more than two categories. This method requires the dependent variable to be categorical and is not suitable for predicting continuous values, where linear regression is more appropriate.

Types of Logistic Regression:

Therearethreemaintypesoflogisticregression:

binary

multinomial

ordinal.

.Binary logistic regression is used when there are two outcomes (yes/no), multinomial is used for three or more categories, and ordinal is used for ordered categories (like ratings). Logistic regression is a classification model that uses input features to predict categorical outputs. It works best when data has low correlation between variables, fewer outliers, and is not too sparse.

LibrariesUsed:

Pandas:

Sklearn:

Seaborn:

Matplotlib:

CONCLUSION:

In this experiment we have studied about the logistic regression model. We ahve performed the classification on the social network.Ads dataset using various liatories of python.

Practical 6

AIM: IMPLEMENT SIMPLE NAVIE BAYESCLASSIFICATIONSALGORITHM. USING PYTHON ON IRIS.CSV DATASET.

REQUIREMENT:

AnacondaInstaller

Windows10OS

JupyterNotebook

THEORY:

Naïve Bayes Classifier:

- Naïve Bayes is a **supervised learning algorithm** used for **classification problems**.
- It is based on **Bayes' Theorem** and works on **probability**.
- It is simple, fast, and effective, especially for **text classification** (spam detection, sentiment analysis).
- It assumes that all features are **independent** of each other.

Why it is called Naïve Bayes:

- **Naïve:** Assumes features are independent.
- **Bayes:** Based on Bayes' Theorem (probability concept).

Advantages:

- Fast and easy to implement
- Works well with small datasets
- Suitable for multi-class problems
- Performs well with categorical data

Disadvantages:

- Assumes feature independence (not always true)
- Cannot handle unseen categories well (zero frequency problem)
- Probability predictions may not always be accurate

Applications of Naïve Bayes:

- Used for **real-time predictions** due to its speed
- Useful for **multi-class classification problems**
- Applied in **spam email filtering** (e.g., Gmail)
- Used in **sentiment analysis** (positive/negative reviews)
- Helps in **recommendation systems**

Types of Naïve Bayes Classifier:

- **Bernoulli Naïve Bayes:** Works with **binary data** (True/False, Yes/No)
- **Multinomial Naïve Bayes:** Used for **text classification** based on word frequency
- **Gaussian Naïve Bayes:** Used for **continuous data** assuming normal distribution

LibrariesUsed:

Pandas:

Sklearn

CONCLUSION:

In this experiment we have studied about Navie bayes algorithm and implemented it an the iris dataset for spam filteration

Practical 7

AIM:

Extract sample document and apply following document preprocessing methods: tokenization, pos tagging, stop words removal, stemming and lemmatization.

REQUIREMENT:

AnacondaInstaller

Windows10OS

Linux

JupyterNotebook

THEORY:

Data Preprocessing in Python:

- Data preprocessing is the process of **cleaning and transforming raw data** before using it in algorithms.
- Raw data is often **unstructured and not suitable** for analysis.
- It helps in converting data into a **clean and usable format**.

Need of Data Preprocessing:

- Improves **accuracy of machine learning models**
- Handles **missing values and inconsistencies**
- Ensures data is in the **correct format for algorithms**
- Helps in applying **multiple ML models on the same dataset**

NLP Techniques in Data Science:

1. Tokenization:

- Splitting text into **words or sentences**

2. Stop Words Removal:

- Removes common words like **"the", "is", "in"**
- Helps reduce **unnecessary data and processing time**

Lemmatization:

- Converts words to their **base/root form with meaning**
- Example: *running → run*
- More **accurate than stemming**

Applications:

- Search engines
- Text analysis systems

Advantages:

- More accurate
- Maintains proper meaning

Disadvantages:

- Slower and more complex

Stemming:

- Reduces words to their **root form (may not be meaningful)**
- Example: *liked, likes → like*

Errors in Stemming:

- **Overstemming:** Different words become same root
- **Understemming:** Similar words remain different

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LibrariesUsed:

NLTK:.

Sklearn

Pandas:

CONCLUSION:

In this experiment we have studied about data preprocessingusing natural language processing technique.

Practical 8

AIM:

Use the inbuilt dataset “titanic” the dataset contains 891 rows and contains informationabout the passengers who boarded the unfortunate titanic ship usethe seaborn liabrary to see if we can find any patterns in the data.

REQUIREMENT:

AnacondaInstaller

Windows10OS

Linux

JupyterNotebook

THEORY:

Data Visualization

- Data visualization is the **graphical representation of data** using charts, graphs, and maps.
- It helps to **understand trends, patterns, and outliers easily**.
- It is very useful in **Big Data** for making better decisions.

Benefits of Data Visualization:

- **Better Analysis:** Helps analyze data like sales and performance easily
- **Quick Action:** Visuals help in making faster decisions
- **Identifying Patterns:** Shows relationships and trends in data
- **Finding Errors:** Helps detect mistakes in data quickly
- **Understanding the Story:** Makes data easy to understand at a glance
- **Business Insights:** Helps in making strategic decisions
- **Latest Trends:** Identifies current trends for better planning

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How Data Visualization Works:

- Converts large amounts of data into **visual formats (charts, graphs, dashboards)**
- Uses tools that connect to **files, databases, APIs, etc.**
- Helps in **analyzing and presenting data effectively**

Features of Good Visualization Tools:

- Interactive charts and dashboards
- Easy connection to different data sources
- Ability to combine data
- Automatic data updates
- Data sharing and exporting options
- Secure access to data

Importance of Real-Time Visualization:

- Helps in making **quick business decisions**
- Handles **large and complex data easily**
- Makes data **easy to understand and explain**

LibrariesUsed:

1. Seaborn
2. matplotlib

CONCLUSION:

In this experiment we have studied about what is data visualization. How data visualization helps us in different ways. We have also used the inbuilt data set i.r., Titanic from the seaborn liabrary to perform data visualization on it.

Practical 9

AIM:

Use the inbuilt dataset ‘titanic’ as used in previous experiment . Plot a box plot for distribution of age with respect to each gender along with the information anout wheather they survived or not.(cloumn name ‘sex’ and ‘age’)

REQUIREMENT:

AnacondaInstaller

Windows10OS

Linux

JupyterNotebook

THEORY:

What is Data Visualization:

- Data visualization is the **graphical representation of data** using charts, graphs, and maps.
- It helps to **understand trends, patterns, and outliers easily**.
- It is important in **Big Data** to analyze large datasets and make decisions.

Uses of Data Visualization:

- Helps to **model complex events**
- Used to visualize things that cannot be seen directly like:
    - Weather patterns
    - Medical data
    - Mathematical relationships

Data Visualization Technique: Box and Whisker Plot (Box Plot):

- Used to show **statistical summary of data**
- Displays:
    - **Minimum and Maximum values (whiskers)**
    - **Quartiles (25%, 50%, 75%)**
- Helps to identify **outliers** (values outside whiskers)
- Shows **Interquartile Range (IQR)** for data spread

Implementation:

- Box plot is available in **Seaborn library**
- Used for **univariate analysis** (one variable)
- Helps to study the effect of one feature on another (e.g., feature vs class)

LibrariesUsed:

Seaborn

Conclusion

in this experiment we have studied about the data visualization technique and implemented data visualization on the built in data set in seaborn laibrary i.e., titanic dataset

Practical 10

AIM:

Download the Iris flower dataset or any other dataset into a DataFrame. (e.g., https://archive.ics.uci.edu/ml/datasets/Iris). Scan the dataset and give the inference as:
1. List down the features and their types (e.g., numeric, nominal) available in the dataset.
2. Create a histogram for each feature in the dataset to illustrate the feature distributions.
3. Create a box plot for each feature in the dataset.
4. Compare distributions and identify outliers.

REQUIREMENT:

AnacondaInstaller

Windows10OS

Linux

JupyterNotebook

THEORY:

What is DataVisualization?

Datavisualizationsimplyreferstotechniquesthatareusedtocommunicate with insights from data through a visual representation. To simplify the thing, you can say that data visualization visually puts data. So we can easily understand it. One has to master different data visualization tools to become effective.

Data visualization is what will help you out. To learn more about data visualization and other visual representation in data science, check out our datascience certifications from recognized universities.

Categorical Data:

- Categorical data represents **categories or labels**
- Types of categorical data:
    - **Nominal:** No specific order (e.g., colors, names)
    - **Ordinal:** Has a specific order (e.g., ranks, grades)

Formats of Categorical Data:

- **Raw Data:** Individual observations
- **Aggregated Data:** Counts of each category
- **Cross-tabulated Data:** Data shown in table form (rows & columns)

Working with Categorical Variables:

- Represented as:
  - **Character vectors**
  - **Factors**

Advantages of Factors:

- Better control over **order of categories**
- Levels remain preserved during **data operations**

Common Functions (R):

- **factor():** Converts data into factor
- **levels():** Shows categories (levels)
- **reorder():** Changes order of levels

LibrariesUsed:

Seaborn:

Pandas:

Numpy

Sklearn:

CONCLUSION:

In this experiment we have studied about the data visualization and performed data visualization on the iris data set in different ways.